

Definition 6. A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is called *L-smooth* if it is continuously differentiable with *L-Lipschitz gradient*. That is, for all $\mathbf{x}, \mathbf{y}$ in the domain,

$$\|\nabla f(\mathbf{x}) - \nabla f(\mathbf{y})\|_2 \leq L\|\mathbf{x} - \mathbf{y}\|_2.$$

Definition 7. A function $f: \mathbb{R}^n \rightarrow \mathbb{R}$ is called *σ -strongly convex* if, for all $\mathbf{x}, \mathbf{y}$ in the domain,

$$f(\mathbf{y}) \geq f(\mathbf{x}) + \nabla f(\mathbf{x})^\top (\mathbf{y} - \mathbf{x}) + \frac{\sigma}{2} \|\mathbf{y} - \mathbf{x}\|_2^2.$$

In particular, recall that strong convexity implies the existence of a unique minimum. Hence, it makes sense to write, for example, $\mathbf{x}^* = \arg \max_{\mathbf{x}} f(\mathbf{x})$ without ambiguity.

C.1 Gradient descent with untied weights

Theorem 8. Assume $\ell: \Theta^n \rightarrow \mathbb{R}$ is σ -strongly convex and *L-smooth* for some $\sigma, L > 0$. Consider ordinary gradient descent with untied weights

$$\boldsymbol{\theta}_i^{(t)} = \boldsymbol{\theta}_i^{(t-1)} - \eta \nabla_{\boldsymbol{\theta}_i} \ell(\boldsymbol{\theta}_1^{(t-1)}, \dots, \boldsymbol{\theta}_i^{(t-1)}) \quad \forall i \in \{1, \dots, n\}.$$

Then, for $\boldsymbol{\theta}_1^*, \dots, \boldsymbol{\theta}_n^* = \arg \min_{\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_n} \ell(\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_n)$, for small enough $\eta > 0$,

$$\|\boldsymbol{\theta}_i^{(t)} - \boldsymbol{\theta}_i^*\|_2^2 \leq \left(1 - \frac{2\eta\sigma L}{\sigma + L}\right)^t \|\boldsymbol{\theta}_i^{(0)} - \boldsymbol{\theta}_i^*\|_2^2 \quad \forall i \in \{1, \dots, n\}.$$

Proof. This is a standard convergence result for gradient descent on strongly convex functions. For example, see [Nesterov \(2018\)](#). $\square$

C.2 Gradient descent with tied weights

Theorem 9. Assume ℓ is σ -strongly convex and *L-smooth*, and consider ordinary gradient descent with tied weights

$$\begin{aligned} \boldsymbol{\theta}_1^{(0)} &= \dots = \boldsymbol{\theta}_n^{(0)} \\ \boldsymbol{\theta}_i^{(t+1)} &= \boldsymbol{\theta}_i^{(t)} + \eta \sum_{i=1}^n \nabla_{\boldsymbol{\theta}_i} \ell(\boldsymbol{\theta}_1, \dots, \boldsymbol{\theta}_n) \quad \forall i \in \{1, \dots, n\}. \end{aligned}$$

There exists $\eta > 0$ such that

$$\|\boldsymbol{\theta}_i^{(t)} - \boldsymbol{\theta}^*\|_2^2 \leq \left(1 - \frac{2\sqrt{n}\eta\sigma L}{\sigma + L}\right)^t \|\boldsymbol{\theta}_i^{(0)} - \boldsymbol{\theta}^*\|_2^2 \quad \forall i \in \{1, \dots, n\}.$$

where $\boldsymbol{\theta}^* = \arg \min_{\boldsymbol{\theta}} \ell(\boldsymbol{\theta}, \dots, \boldsymbol{\theta})$.

Proof. This is again the standard convergence result, now applied to descent with step size $\sqrt{n}\eta$ on the σ -strongly convex *L-smooth* function $\boldsymbol{\theta} \mapsto \ell(\boldsymbol{\theta}/\sqrt{n}, \dots, \boldsymbol{\theta}/\sqrt{n})$. Alternatively, one may think of this as projected gradient descent constrained to the subspace $\boldsymbol{\theta}_1 = \dots = \boldsymbol{\theta}_n$ with a step size of $\sqrt{n}\eta$. Projected gradient descent inherits the same convergence properties as unconstrained gradient descent ([Beck, 2017](#)). $\square$

C.3 Myopic descent with untied weights

Theorem 10. Assume each of the $\ell_1, \dots, \ell_n$ are σ -strongly convex and *L-smooth*. Consider myopic gradient descent with untied weights

$$\boldsymbol{\theta}_i^{(t)} = \boldsymbol{\theta}_i^{(t-1)} - \eta \nabla_{\boldsymbol{\theta}_i} \ell_i(\boldsymbol{\theta}_1^{(t-1)}, \dots, \boldsymbol{\theta}_i^{(t-1)}) \quad \forall i \in \{1, \dots, n\}.$$

There exists $\eta > 0$ such that $\theta_i \xrightarrow{t \rightarrow \infty} \tilde{\theta}_i$ for all i , where

$$\begin{aligned}\tilde{\theta}_1 &= \arg \min_{\theta_1} \ell_1(\theta_1) \\ \tilde{\theta}_2 &= \arg \min_{\theta_2} \ell_2(\tilde{\theta}_1, \theta_2) \\ &\dots \\ \tilde{\theta}_n &= \arg \min_{\theta_n} \ell_n(\tilde{\theta}_1, \tilde{\theta}_2, \dots, \tilde{\theta}_{n-1}, \theta_n).\end{aligned}$$

Proof. Let $\tilde{\theta}_1, \dots, \tilde{\theta}_n$ be as in the theorem statement. We proceed by induction. For the base case, note that the myopic descent iterates for $\theta_1^{(t)}$ are independent of $\theta_j^{(t)}$ for $j > 1$. Thus, the standard convergence theorem gives that $\theta_1^{(t)} \rightarrow \tilde{\theta}_1$ as $t \rightarrow \infty$.

Now, assume $\theta_i^{(t)} \xrightarrow{t \rightarrow \infty} \tilde{\theta}_i$ for all $i < k$. Thus, for any $\varepsilon > 0$, for sufficiently large t ,

$$\|(\theta_i^{(t)})_{i=1}^{k-1} - (\tilde{\theta}_i)_{i=1}^{k-1}\|_2 < \varepsilon.$$

Hence, since ℓ_k is L -smooth, for any θ_k ,

$$\|\nabla_{\theta_k} \ell_k(\theta_1^{(t)}, \dots, \theta_{k-1}^{(t)}, \theta_k) - \nabla_{\theta_k} \ell_k(\tilde{\theta}_1, \dots, \tilde{\theta}_{k-1}, \theta_k)\|_2 < L\varepsilon.$$

Expanding and rearranging,

$$\langle \nabla_{\theta_k} \ell_k(\theta_1^{(t)}, \dots, \theta_{k-1}^{(t)}, \theta_k), \nabla_{\theta_k} \ell_k(\tilde{\theta}_1, \dots, \tilde{\theta}_{k-1}, \theta_k) \rangle \geq \frac{1}{2} \|\nabla_{\theta_k} \ell_k(\tilde{\theta}_1, \dots, \tilde{\theta}_{k-1}, \theta_k)\|_2^2 - \frac{1}{2} L^2 \varepsilon^2$$

is bounded away from zero as long as, say,

$$\|\nabla_{\theta_k} \ell_k(\tilde{\theta}_1, \dots, \tilde{\theta}_{k-1}, \theta_k)\|_2 \geq \sigma \|\theta_k - \tilde{\theta}_k\|_2 > (L+1)\varepsilon,$$

using the σ -strong convexity of ℓ_k . That is, as long as $\|\theta_k - \tilde{\theta}_k\|_2 > \frac{(L+1)\varepsilon}{\sigma}$, it is guaranteed that $-\nabla_{\theta_k} \ell_k(\theta_1^{(t)}, \dots, \theta_{k-1}^{(t)}, \theta_k)$ is a descent direction for $\ell_k(\tilde{\theta}_1, \dots, \tilde{\theta}_{k-1}, \theta_k)$. This is a sufficient condition for θ_k to converge to a $\frac{(L+1)\varepsilon}{\sigma}$ -neighborhood of $\tilde{\theta}_k$ given a small enough step size (Beck, 2017). Since $\varepsilon > 0$ is arbitrary, this completes the inductive step. $\square$

C.4 Myopic descent with tied weights

Definition 11. A function $f(x, y): \mathbb{R}^a \times \mathbb{R}^a \rightarrow \mathbb{R}$ with continuous second derivatives is ρ -forward-biased if, for all $y \in \mathbb{R}^a$ and $x = y$,

$$H_{y,y}f(x, y) + H_{x,y}f(x, y) \succ 0$$

and

$$\kappa(H_{y,y}f(x, y) + H_{x,y}f(x, y)) < \rho,$$

where κ is the condition number, and we write the Hessian of f as a block matrix:

$$Hf(x, y) = \begin{bmatrix} H_{x,x}f(x, y) & H_{x,y}f(x, y) \\ H_{y,x}f(x, y) & H_{y,y}f(x, y) \end{bmatrix} = \begin{bmatrix} \left(\frac{\partial f(x, y)}{\partial x_i \partial x_j} \right)_{\substack{1 \leq i \leq a \\ 1 \leq j \leq a}} & \left(\frac{\partial f(x, y)}{\partial x_i \partial y_j} \right)_{\substack{1 \leq i \leq a \\ 1 \leq j \leq b}} \\ \left(\frac{\partial f(x, y)}{\partial y_i \partial x_j} \right)_{\substack{1 \leq i \leq b \\ 1 \leq j \leq a}} & \left(\frac{\partial f(x, y)}{\partial y_i \partial y_j} \right)_{\substack{1 \leq i \leq b \\ 1 \leq j \leq b}} \end{bmatrix}.$$

Lemma 12. If $A \in \mathbb{R}^{a \times a}$ is positive definite with $\kappa(A) < \rho$, then there exists some $\eta > 0$ such that $I - \eta A$ is a $(1 - \rho^{-1})$ -contraction. Explicitly, for any $y \in \mathbb{R}^a$,

$$\|(I - \eta A)y\|_2 \leq (1 - \rho^{-1})\|y\|_2.$$

Proof. Set $\eta = 1/\lambda_{\max}(\mathbf{A})$. Then $\mathbf{I} - \eta\mathbf{A} \succeq 0$ with

$$\lambda_{\max}(\mathbf{I} - \eta\mathbf{A}) = 1 - \eta\lambda_{\min}(\mathbf{A}) = 1 - \frac{\lambda_{\min}(\mathbf{A})}{\lambda_{\max}(\mathbf{A})} < 1 - \rho^{-1}.$$

It immediately follows that $\mathbf{I} - \eta\mathbf{A}$ is a $(1 - \rho^{-1})$ -contraction. $\square$

Theorem 13. Let $f(\mathbf{x}, \mathbf{y}) : \mathbb{R}^a \times \mathbb{R}^a \rightarrow \mathbb{R}$ be ρ -forward biased, σ -strongly convex, and L -smooth with continuous second derivatives for some $\rho, \sigma, L > 0$. Then, there exists $\tilde{\mathbf{y}} \in \mathbb{R}^a$ such that

$$\tilde{\mathbf{y}} = \arg \min_{\mathbf{y} \in \mathbb{R}^a} f(\tilde{\mathbf{y}}, \mathbf{y}). \quad (4)$$

Further, for some step size $\eta > 0$, the iterates of myopic gradient descent with tied weights

$$\mathbf{y}^{(t+1)} = \mathbf{y}^{(t)} - \eta \nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})|_{\mathbf{x}=\mathbf{y}=\mathbf{y}^{(t)}}$$

converge to $\tilde{\mathbf{y}} \in \mathbb{R}^a$ satisfying (4). We call such $\tilde{\mathbf{y}}$ a myopic solution.

We then recover the original sequential modeling setting by defining $f(\tilde{\boldsymbol{\theta}}, \boldsymbol{\theta}) := \sum_{i=1}^n \ell_i(\tilde{\boldsymbol{\theta}}, \dots, \tilde{\boldsymbol{\theta}}, \boldsymbol{\theta})$.

Proof. First, note that if myopic descent does converge, it must be to a point $\tilde{\mathbf{y}}$ such that $\tilde{\mathbf{y}} = \arg \min_{\mathbf{y}} f(\tilde{\mathbf{y}}, \mathbf{y})$. Indeed, at convergence we must have $\nabla_{\mathbf{y}} f(\tilde{\mathbf{y}}, \mathbf{y})|_{\mathbf{y}=\tilde{\mathbf{y}}} = 0$, so strong convexity tells us that $\tilde{\mathbf{y}}$ is optimal. Thus, if we establish that myopic descent with small enough step size $\eta > 0$ converges to some $\tilde{\mathbf{y}}$, we automatically get the existence of a myopic solution. For this, it suffices to show that the gradient descent step

$$g_{\eta}(\mathbf{y}) = \mathbf{y} - \eta \nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})|_{\mathbf{x}=\mathbf{y}}$$

is strictly contractive, so that iterates of g_{η} converge to a fixed point.

Consider arbitrary $\mathbf{y}$ and $\mathbf{y}'$. Then, by chain rule and the mean value theorem applied to the map $\mathbf{y}' \mapsto \nabla_{\mathbf{y}} f(\mathbf{x}, \mathbf{y})|_{\mathbf{x}=\mathbf{y}=\mathbf{y}'}$,

$$\nabla_{\mathbf{y}} f(\mathbf{y}', \mathbf{y}') - \nabla_{\mathbf{y}} f(\mathbf{y}, \mathbf{y}) = (\mathbf{H}_{\mathbf{x}, \mathbf{y}} f(\mathbf{y}'', \mathbf{y}'') + \mathbf{H}_{\mathbf{y}, \mathbf{y}} f(\mathbf{y}'', \mathbf{y}''))(\mathbf{y}' - \mathbf{y}) \quad (5)$$

for some $\mathbf{y}'' \in [\mathbf{y}, \mathbf{y}']$. Using the definition of g_{η} and substituting in (5),

$$\begin{aligned} \|g_{\eta}(\mathbf{y}') - g_{\eta}(\mathbf{y})\|_2^2 &= \|(\mathbf{y}' - \mathbf{y}) - \eta(\nabla_{\mathbf{y}} f(\mathbf{y}', \mathbf{y}') - \nabla_{\mathbf{y}} f(\mathbf{y}, \mathbf{y}))\|_2^2 \\ &= \|(I - (\mathbf{H}_{\mathbf{x}, \mathbf{y}} f(\mathbf{y}'', \mathbf{y}'') + \mathbf{H}_{\mathbf{y}, \mathbf{y}} f(\mathbf{y}'', \mathbf{y}'')))(\mathbf{y}' - \mathbf{y})\|_2^2 \\ &\leq (1 - \rho^{-1})(\mathbf{y}' - \mathbf{y}), \end{aligned}$$

where the last line is by ρ -forward bias and Lemma 12. That is, g_{η} is $(1 - \rho^{-1})$ -contractive, completing the proof. $\square$

C.5 Properties of sine

Lemma 14. Let $\mathbf{x} \sim \mathcal{N}(0, 1)$. Then

$$\text{Var}(\sin(b\mathbf{x})) = \frac{1}{2} - \frac{e^{2b^2}}{2}.$$

Proof.

$$\text{Var}(\sin(b\mathbf{x})) = (2\pi)^{-1/2} \int_{-\infty}^{\infty} \sin^2(bx) e^{-x^2/2} dx = \frac{1}{2} - \frac{e^{2b^2}}{2}.$$

$\square$

Lemma 15. Let $\mathbf{x} \sim \mathcal{N}(0, 1)$. Then

$$\rho(\mathbf{x}, \sin(b\mathbf{x})) = \frac{2be^{3b^2/2}}{e^{2b^2} - 1} \in O(e^{-b^2/2}),$$

where ρ is the Pearson correlation coefficient.